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Algorithms

RNA secondary structural alignment with conditional random fields

Kengo Sato* and Yasubumi Sakakibara

Department of Biosciences and Informatics, Keio University 3-14-1 Hiyoshi, Kohoku-ku, Yokohama 223-8522, Japan

ABSTRACT

Motivation: The computational identification of non-coding RNA regions on the genome is currently receiving much attention. However, it is essentially harder than gene-finding problems for protein-coding regions because non-coding RNA sequences do not have strong statistical signals. Since comparative sequence analysis is effective for non-coding RNA detection, efficient computational methods are expected for structural alignment of RNA sequences. Several methods have been proposed to accomplish the structural alignment tasks for RNA sequences, and we found that one of the most important points is to estimate an accurate score matrix for calculating structural alignments.

Results: We propose a novel approach for RNA structural alignment based on conditional random fields (CRFs). Our approach has some specific features compared with previous methods in the sense that the parameters for structural alignment are estimated such that the model can most probably discriminate between correct alignments and incorrect alignments, and has the generalization ability so that a satisfiable score matrix can be obtained even with a small number of sample data without overfitting. Experimental results clearly show that the parameter estimation with CRFs can outperform all the other existing methods for structural alignments of RNA sequences. Furthermore, structural alignment search based on CRFs is more accurate for predicting non-coding RNA regions than the other scoring methods. These experimental results strongly support our discriminative method employing CRFs to estimate the score matrix parameters.

Availability: The program which is implemented in C++ is available at http://phmmts.dna.bio.keio.ac.jp/ under the GNU public license.

Contact: satoken@bio.keio.ac.jp

1 INTRODUCTION

The computational identification of non-coding RNA regions on the genome is currently receiving much attention (Eddy, 2001; Schattner, 2002). However, it is essentially harder than gene-finding problems for protein-coding regions because non-coding RNA sequences do not have strong statistical signals, and there are no general finding algorithms yet. Since comparative sequence analysis is effective for non-coding RNA detection, efficient computational methods are expected for structural alignment of RNA sequences. By structural alignment, we mean a pairwise alignment to align an unfolded RNA sequence into an RNA sequence of known secondary structure. Several methods including stochastic context-free grammars (SCFGs) (Sakakibara *et al*., 1994; Eddy and Durbin, 1994), FOLDALIGN (Gorodkin *et al*., 1997), pair hidden Markov models on trees structures (PHMMTSs) (Sakakibara, 2003), RSEARCH (Klein and Eddy, 2003) and PSTAGs (Matsui *et al*., 2005) have been proposed to accomplish the structural alignment tasks for RNA sequences.

PHMMTSs proposed by Sakakibara (2003) is one of algorithms for structural alignment of a binary tree and a sequence, and has been applied to aligning RNA secondary structures, i.e. a pairwise alignment to align an unfolded RNA sequence into an RNA sequence of known secondary structure. Furthermore, PHMMTSs can used for identifying non-coding RNA regions in an unfolded RNA sequence by aligning it into non-coding RNAs of known secondary structure. In order to calculate structural alignments of RNA sequences, PHMMTS requires some parameters, such as the emission probability of base pairs and the state transition probability, which have much effect on the performance of alignments. Thus, estimating these parameters is one of the most crucial problems for RNA structural alignment.

There are some related works to estimating the parameters for aligning RNA secondary structures (Gorodkin *et al*., 1997; Klein and Eddy, 2003). Gorodkin *et al*. (1997) have developed a score matrix for matching a pair of base pairs, which is defined to be the sum of matrices for alignment scores and base pairing scores so that both of them can be independent of each other, and have been found to work well on their algorithm FOLDALIGN, although each parameter seems to be determined *ad hoc*. Klein and Eddy (2003) have proposed a ribosomal RNA substitution matrix, called RIBOSUM, which is based on an analogous method to the BLOSUM matrices (Henikoff and Henikoff, 1992). However, since the RIBOSUM matrix is based on the maximum-likelihood (ML) estimation by relative frequencies of RNA mutations, to avoid overfitting requires a large number of high quality structure-annotated alignments, although there are not so many. Therefore, more effective methods for estimating the parameters for RNA structural alignment should be developed.

In this paper, we propose a novel approach for RNA structural alignment based on conditional random fields (CRFs). CRFs proposed by Lafferty *et al*. (2001) have several advantages over traditional HMMs and stochastic grammars because CRFs can build discriminative models with flexible state transitions from annotated training data. Therefore, CRFs have applied to various tasks on segmenting and labeling sequence data (Sha and Pereira, 2003; Settles, 2004; McDonald and Pereira, 2004; Liu *et al*., 2004). Hence, our approach has some specific features compared with previous methods

*To whom correspondence should be addressed.

in the sense that the parameters for structural alignment are estimated such that the model can most probably discriminate between correct alignments and incorrect alignments, and has the generalization ability so that a satisfiable score matrix can be obtained even with a small number of sample data without overfitting.

Experimental results clearly show that the parameter estimation with CRFs can outperform all the other existing methods for structural alignments of RNA sequences. Furthermore, structural alignment search based on CRFs is more accurate for predicting non-coding RNA regions than the other scoring methods. These experimental results strongly support our discriminative method employing CRFs to estimate the score matrix parameters.

2 METHODS

2.1 Structural alignment

Aligning structured data computes a similarity score between two (or more) structured data, such as sequences, trees and graphs. An alignment is a set of operations in which some null (gap) nodes are inserted into the structures such that two resulting structures are identical to each other. As alignment operations (state), 'match' (M), 'deletion' (D) and 'insertion' (I) are usually used, and each of them is associated with emission scores/probabilities of symbols and state transition scores/probabilities between these operations. In the simple linear sequence alignment, any state transits from a single state to a single state. However, more complex case, e.g. alignment of rooted binary trees, may have some bifurcation state transitions, each of which bifurcates into two states.

Let $a = a_1, a_2, \ldots, a_l$ be an alignment of given rooted binary trees t^1 and t^2, and a_1 be a root of a. Then, the probability of a is calculated by the following recurrence equation.

$$P(a[i]) = \begin{cases} \sigma(a_i) \quad (a_i \text{ has no child}) \\ \tau(a_i \to a_j)\sigma(a_i)P(a[j]) \\ \quad (a_i \text{ has one child } a_j) \\ \tau(a_i \to a_j\, a_k)\sigma(a_i)P(a[j])P(a[k]) \\ \quad (a_i \text{ has two children } a_j \text{ and } a_k), \end{cases}$$

where $a[i]$ is a subtree of a whose root is a_i, $\sigma(a_i)$ is an emission probability of a_i, $\tau(a_i \to a_j)$ is a state transition probability from a_i to a_j, and $\tau(a_i \to a_j\, a_k)$ is a state transition probability that a_i bifurcates into a_j and a_k. Thus, the probability of an alignment a of given t^1 and t^2 is defined as $P(t^1, t^2, a) = P(a[1])$. The optimal alignment $\hat{a}$ of t^1 and t^2 satisfies

$$\begin{aligned} \hat{a} &= \operatorname*{arg\,max}_{a \in \mathcal{A}(t^1,t^2)} P\big(a|t^1, t^2\big) \\ &= \operatorname*{arg\,max}_{a \in \mathcal{A}\left(t^1,t^2\right)} \frac{P(t^1, t^2, a)}{\sum_{a' \in \mathcal{A}(t^1,t^2)} P\big(t^1, t^2, a'\big)} \\ &= \operatorname*{arg\,max}_{a \in \mathcal{A}(t^1,t^2)} P\big(t^1, t^2, a\big), \end{aligned} \tag{1}$$

where $\mathcal{A}(t^1, t^2)$ is a set of all possible alignments of t^1 and t^2, and $\hat{a}$ can be obtained by dynamic programming techniques.

If t^2 is not a binary tree but a sequence, the optimal alignment of a binary tree t^1 and a sequence t^2 can be found by synchronizing dynamic programming for Equation (1) with the CYK algorithm (Durbin *et al.*, 1998) for t^2, which is one of the parsing algorithms of stochastic context-free grammars (SCFGs).

PHMMTSs proposed by Sakakibara (2003) is one of algorithms for structural alignment of a binary tree and a sequence, and has been applied for aligning RNA secondary structures, that is, a pairwise alignment to align an unfolded RNA sequence into an RNA sequence of known secondary structure. Furthermore, PHMMTSs can be used for identifying non-coding RNA regions in an unfolded RNA sequence by aligning it into non-coding RNAs of known secondary structure.

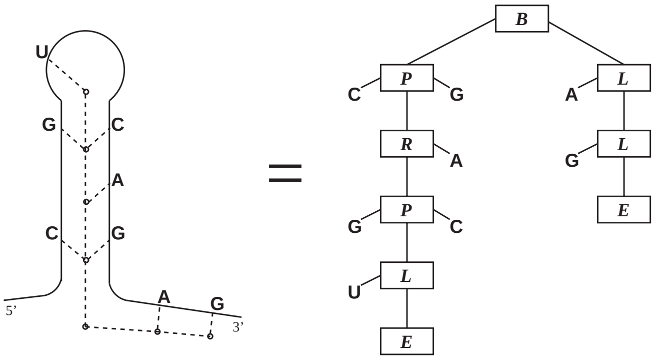

Fig. 1. A binary tree representing a folded RNA sequence. The left panel shows a folded RNA sequence with secondary structure and the right panel shows its binary tree representation.

Let s be an unfolded RNA sequence of the length n, s_i be the i-th symbol of s, and $s[i, j]$ be a subsequence of s from the i-th symbol to the $(j-1)$-th symbol. Let t be a binary tree which represents a folded RNA sequence of known secondary structure as shown in Figure 1, t_u be a node of t, specially, t_1 be the root node of t, and $t[u]$ is a subtree of t whose root is t_u. Each node of t is one of five types of the node: B, P, L, R and E, which represent bifurcation nodes, base pair nodes, left-base nodes, right-base nodes and end nodes, respectively.

For finding the optimal structural alignment of an unfolded RNA sequence s and a folded RNA structure t, Sakakibara (2003) has presented the following recurrence equations.

$$\begin{aligned} &P_M(s[i,j], t[u]) \\ &\quad = \begin{cases} (\text{if } t_u = B) \quad \max\limits_{\substack{X,Y \in \{M,I,D\}, \\ i<l\le j}} \sigma_M(t_u)\tau(M \to XY) \\ \qquad\qquad \times P_X(s[i,l-1], t[v]) P_Y(s[l,j], t[w]) \\ (\text{if } t_u = P) \\ \quad \max\limits_{X \in \{M,I,D\}} \sigma_M(s_i, s_j, t_u)\tau(M \to X) P_X(s[i+1, j-1], t[v]) \\ (\text{if } t_u = L) \\ \quad \max\limits_{X \in \{M,I,D\}} \sigma_M(s_i, \varepsilon, t_u)\tau(M \to X) P_X(s[i+1, j], t[v]) \\ (\text{if } t_u = R) \\ \quad \max\limits_{X \in \{M,I,D\}} \sigma_M(\varepsilon, s_j, t_u)\tau(M \to X) P_X(s[i, j-1], t[v]) \\ (\text{if } t_u = E) \\ \quad 1 \quad \text{for } s[i,j] = \varepsilon, \quad 0 \quad \text{for the others,} \end{cases} \end{aligned}$$

$$\begin{aligned} &P_I(s[i,j], t[u]) \\ &\quad = \max_{\substack{X \in \{M,I,D\}, \\ i<l\le j}} \begin{cases} \sigma_I(\theta)\tau(I \to XI) \\ \quad \times P_X(s[i,l-1], t[u]) P_I(s[l,j], \theta) \\ \sigma_I(\theta)\tau(I \to IX) \\ \quad \times P_I(s[i,l-1], \theta) P_X(s[l,j], t[u]) \\ \sigma_I(s_i, s_j, \theta)\tau(I \to X) P_X(s[i+1, j-1], t[u]) \\ \sigma_I(s_i, \varepsilon, \theta)\tau(I \to X) P_X(s[i+1, j], t[u]) \\ \sigma_I(\varepsilon, s_j, \theta)\tau(I \to X) P_X(s[i, j-1], t[u]), \end{cases} \end{aligned} \tag{2}$$

$$\begin{aligned} &P_D(s[i,j], t[u]) \\ &\quad = \begin{cases} (\text{if } t_u = B) \quad \max\limits_{X \in \{M,I,D\}} \begin{cases} \sigma_D(t_u)\tau(D \to XD) \\ \quad \times P_X(s[i,j], t[v]) P_D(\varepsilon, t[w]) \\ \sigma_D(t_u)\tau(D \to DX) \\ \quad \times P_X(\varepsilon, t[v]) P_D(s[i,j], t[w]) \end{cases} \\ (\text{if } t_u = P, L, R) \quad \max\limits_{X \in \{M,I,D\}} \sigma_D(\varepsilon, \varepsilon, t_u)\tau(D \to X) \\ \qquad\qquad \times P_X(s[i,j], t[v]), \end{cases} \end{aligned}$$

where ε is the empty sequence, θ is the empty tree, σ_X is an emission probability on state X and τ is a state transition probability. Each t_u has either two children t_v and t_w for B or a child t_v for the others. The probability of the optimal alignment of s and t can be calculated by $\max_{X\in\{M,I,D\}} P_X(s[1,n+1],t[1])$. The optimal secondary structure of s can be retrieved by a traceback of the dynamic programming matrix.

Calculating $P_X(s[i,j],t[u])$ requires some parameters, such as the emission probability σ_X and the state transition probability τ, which have much effect on the performance of alignments. Therefore, estimating these parameters is one of the most crucial problems for RNA structural alignment, and an effective method should be developed.

2.2 Conditional random fields

We propose a novel approach for RNA structural alignment based on conditional random fields (CRFs). CRFs proposed by Lafferty *et al*. (2001) have several advantages over traditional HMMs and stochastic grammars because CRFs can build discriminative models with flexible state transitions from annotated training data. Therefore, CRFs have applied to various tasks on segmenting and labeling sequence data (Sha and Pereira, 2003; Settles, 2004; McDonald and Pereira, 2004; Liu *et al*., 2004). CRFs can be applied not only to sequence data, but also to more complex structured data, such as trees and graphs. We focus here on CRFs for structural alignment of RNA secondary structures, although the notion can be more general (Lafferty *et al*., 2001).

Given an unfolded RNA sequence $s = s_1,\ldots,s_n$ and a binary tree $t = t_1,\ldots,t_m$ for a folded RNA sequence of known secondary structure, we define a graph $G_{s,t} = (V_{s,t}, E_{s,t})$, where $V_{s,t} = \{\langle X,i,j,u\rangle | X \in \{M,I,D\}, 1 \le i \le j \le n+1, 1 \le u \le m\}$ is a set of vertices, and $E_{s,t}$ is a set of edges (x,y) and (x,y,z) for $x,y,z \in V_{s,t}$ such that each of them satisfies one of the conditions shown in Table 1. An alignment $a = a_1,\ldots,a_l$ ($a_k \in V_{s,t}, 1 \le k \le l$) is a subtree of $G_{s,t}$ rooted by $a_1 = \langle X,1,n+1,1\rangle$ for $X \in \{M,I,D\}$ which covers s and t. CRFs for structural alignment define the conditional probability distribution $P(a|s,t)$ of an alignment a given an unfolded sequence s and a folded structure t.

An alignment a is specified by a vector $\boldsymbol{f}$ of local features and a corresponding weight vector $\boldsymbol{\lambda}$. We can design elements of $\boldsymbol{f}$ based on various properties of graph, vertices and edges. In our method, each element of a local feature vector $\boldsymbol{f}$ is either an emission feature σ or a state transition feature τ.

$$\sigma_{s_i,s_j,t_u}(a_p,s,t) = \begin{cases} 1 & (a_p \text{ emits } s_i, s_j \text{ and } t_u) \\ 0 & (\text{the others}), \end{cases}$$

$$\tau_{XY}(a_p,a_q,s,t) = \begin{cases} 1 & \left(\begin{array}{l} (a_p,a_q) \in E_{s,t}, \\ \text{the state of } a_p \text{ is } X \text{ and} \\ \text{the state of } a_q \text{ is } Y \end{array} \right) \\ 0 & (\text{the others}), \end{cases}$$

$$\tau_{XYZ}(a_p,a_q,a_r,s,t) = \begin{cases} 1 & \left(\begin{array}{l} (a_p,a_q,a_r) \in E_{s,t}, \\ \text{the state of } a_p \text{ is } X, \\ \text{the state of } a_q \text{ is } Y \text{ and} \\ \text{the state of } a_r \text{ is } Z \end{array} \right) \\ 0 & (\text{the others}), \end{cases}$$

where $1 \le p,q,r \le l$. To make the notation more uniform, we also write each element of a local feature vector $\boldsymbol{f}$ as

$$f(a_p,a_q,a_r,s,t) = \begin{cases} (\text{for emission}) & \sigma_{s_i,s_j,t_u}(a_p,s,t) \\ (\text{for state transition}) & \tau_{XY}(a_p,a_q,s,t) \\ (\text{for state transition with bifurcation}) & \tau_{XYZ}(a_p,a_q,a_r,s,t) \end{cases}$$

for each emission and state transition feature.

The CRF's global feature vector for an unfolded sequence s, a folded sequence t and a structural alignment a is given by

$$\boldsymbol{F}(a,s,t) = \sum_{1\le p,q,r\le l} \boldsymbol{f}(a_p,a_q,a_r,s,t).$$

Table 1. Allowable edges on the graph of structural alignment

x	Type of t_u	y	z
$\langle M,i,j,u\rangle$	B	$\langle X,i,l-1,v\rangle$	$\langle Y,l,j,w\rangle$
	P	$\langle X,i+1,j-1,v\rangle$	—
	L	$\langle X,i+1,j,v\rangle$	—
	R	$\langle X,i,j-1,v\rangle$	—
$\langle I,i,j,u\rangle$	B	$\langle X,i,l-1,u\rangle$	$\langle I,l,j,0\rangle$
	B	$\langle I,i,l-1,0\rangle$	$\langle X,l,j,u\rangle$
	P	$\langle X,i+1,l-1,u\rangle$	—
	L	$\langle X,i+1,l,u\rangle$	—
	R	$\langle X,i,l-1,u\rangle$	—
$\langle D,i,j,u\rangle$	B	$\langle X,i,j,v\rangle$	$\langle D,j,j,w\rangle$
	B	$\langle D,i,i,v\rangle$	$\langle X,i,j,w\rangle$
	P,L,R	$\langle X,i,j,v\rangle$	—

Each t_u has either two children t_v and t_w for B or a child t_v for the others. Each of X,Y is one of the states M,I,D. A vertex whose last index is 0 means the empty tree θ.

The conditional probability distribution defined by the CRF is then

$$P_{\boldsymbol{\lambda}}(a|s,t) = \frac{\exp \boldsymbol{\lambda}\cdot\boldsymbol{F}(a,s,t)}{Z_{\boldsymbol{\lambda}}(s,t)}, \tag{3}$$

where

$$Z_{\boldsymbol{\lambda}}(s,t) = \sum_{a\in\mathcal{A}(s,t)} \exp \boldsymbol{\lambda}\cdot\boldsymbol{F}(a,s,t),$$

and $\mathcal{A}(s,t)$ is a set of all possible alignments of s and t.

The optimal structural alignment of input s and t is calculated by

$$\begin{aligned} \hat{a} &= \operatorname*{arg\,max}_{a\in\mathcal{A}(s,t)} P_{\boldsymbol{\lambda}}(a|s,t) \\ &= \operatorname*{arg\,max}_{a\in\mathcal{A}(s,t)} \boldsymbol{\lambda}\cdot\boldsymbol{F}(a,s,t). \end{aligned} \tag{4}$$

Since $\boldsymbol{\lambda}\cdot\boldsymbol{F}(a,s,t)$ can decompose into a sum of terms for unary and binary trees of states, the optimal alignment $\hat{a}$ can be obtained by the dynamic programming which is identical to PHMMTSs using Equation (2). The corresponding elements of $\boldsymbol{\lambda}$ to σ_{s_i,s_j,t_u}, τ_{XY} and τ_{XYZ} play the same roles to the logarithms of the probabilities σ_X and τ in Equation (2).

The weight vector $\boldsymbol{\lambda}$ can be trained by maximizing the log-likelihood of given training set $\{(a^k,s^k,t^k)\}_{k=1}^N$.

$$\begin{aligned} \mathcal{L}_{\boldsymbol{\lambda}} &= \sum_k \log P_{\boldsymbol{\lambda}}(a^k|s^k,t^k) \\ &= \sum_k \left\{\boldsymbol{\lambda}\cdot\boldsymbol{F}(a^k,s^k,t^k) - \log Z_{\boldsymbol{\lambda}}(s^k,t^k)\right\}. \end{aligned} \tag{5}$$

Because of convexity of $\mathcal{L}_{\boldsymbol{\lambda}}$, $\boldsymbol{\lambda}$ which maximizes $\mathcal{L}_{\boldsymbol{\lambda}}$ can be found at the zero of its gradient

$$\nabla\mathcal{L}_{\boldsymbol{\lambda}} = \sum_k \left\{\boldsymbol{F}(a^k,s^k,t^k) - E_{s^k,t^k}[\boldsymbol{F}]\right\},$$

where

$$E_{s,t}[\boldsymbol{F}] = \sum_{a\in\mathcal{A}(s,t)} P_{\boldsymbol{\lambda}}(a|s,t)\boldsymbol{F}(a,s,t).$$

The expectation $E_{s,t}[\boldsymbol{F}]$ can be computed efficiently by using a variant of the inside–outside algorithm. Let $\alpha(x)$ be an inside probability of $x = \langle X,i,j,u\rangle \in V_{s,t}$ which means structural alignment of $s[i,j]$ and $t[u]$ on state X, and $\beta(x)$ be an outside probability of x. Detailed definitions of $\alpha(x)$ and $\beta(x)$ are omitted here, but can be introduced from Equation (2) in the same manner as the inside algorithm and the outside algorithm of

SCFGs (Durbin *et al*., 1998). Then, $E_{s,t}[\boldsymbol{F}]$ can be calculated by the following equations:

$$E_{s,t}[\boldsymbol{F}] = \sum_{x,y,z \in V_{s,t}} \frac{\gamma(x,y,z,s,t)\boldsymbol{f}(x,y,z,s,t)}{Z_{\boldsymbol{\lambda}}(s,t)},$$

where

$$Z_{\boldsymbol{\lambda}}(s,t) = \sum_{X \in \{M,I,D\}} \alpha(\langle X, 1, n+1, 1\rangle),$$

$$\gamma(x,y,z,s,t) = \begin{cases} \text{(for state transition with bifurcation)} \\ \quad \beta(x)\alpha(y)\alpha(z)\exp \boldsymbol{\lambda} \cdot \boldsymbol{f}(x,y,z,s,t) \\ \text{(the others)} \\ \quad \beta(x)\alpha(y)\exp \boldsymbol{\lambda} \cdot \boldsymbol{f}(x,y,z,s,t). \end{cases}$$

To avoid overfitting, we penalize the likelihood with Gaussian prior (Chen and Rosenfeld, 1999):

$$\mathcal{L}'_{\boldsymbol{\lambda}} = \sum_k \left\{\boldsymbol{\lambda} \cdot \boldsymbol{F}(a^k, s^k, t^k) - \log Z_{\boldsymbol{\lambda}}(s^k, t^k)\right\} - \frac{||\boldsymbol{\lambda}||^2}{2\sigma^2}$$

with its gradient

$$\nabla\mathcal{L}'_{\boldsymbol{\lambda}} = \sum_k \left\{\boldsymbol{F}(a^k, s^k, t^k) - E_{s^k,t^k}[\boldsymbol{F}]\right\} - \frac{\boldsymbol{\lambda}}{\sigma^2}.$$

In our experiments, we fixed $\sigma = 1$.

To solve the optimal $\boldsymbol{\lambda}$, we use the limited-memory quasi-Newton method (L-BFGS) (Liu and Nocedal, 1989) which is one of general purpose convex optimization algorithms.

3 EXPERIMENTAL RESULTS AND DISCUSSION

To confirm our method, we have done some experiments. A dataset used for training and test in our experiments is extracted from ‘Rfam’ RNA family database (Griffiths-Jones *et al*., 2003) http://www.sanger.ac.uk/Software/Rfam/, which provides two alignment sets; a ‘SEED’ set of high quality alignments which are obtained manually, and a ‘FULL’ set of alignments generated by the covariance model (CM) method (Eddy and Durbin, 1994). We randomly extracted structurally aligned pairs of RNA sequences as the dataset from four families in Rfam SEED because these families have enough number of alignments which are biologically plausible. The summary of the dataset is shown in Table 2.

We have trained the parameters of the weight vector $\boldsymbol{\lambda}$ corresponding to the emission features σ and the state transition features τ for structural alignment according to CRF’s training method described in Section 2.2, in which 400 pairs in the dataset are used as the training data. Figure 2 shows the parameters trained by CRF. Then, according to Equation (4), we have aligned 100 pairs of RNA sequences in the rest of the dataset with trained parameters; one of each pair is used as an unfolded RNA sequence, and aligned into the other using its annotations of secondary structures.

The result of these has been evaluated by specificity and sensitivity of predicted base pairs, the rate of correctly predicted base pairs to all of predicted base pairs and the rate of correctly predicted base pairs to all of the annotated base pairs in the dataset.

Table 3 shows the accuracy of predicting base pairs by structural alignment for each family. Here, CRF is compared with the other scoring methods: EM by the expectation–maximization (EM) algorithm, RIBOSUM by (Klein and Eddy, 2003), and FOLDALIGN by (Gorodkin *et al*., 1997). The result clearly shows that CRF can outperform all the other existing methods for structural alignment of RNA sequences.

Table 2. The summary of the dataset used in our experiments

Accession	Len.	Homo. (%)	Description
RF00001	116.8	70.1	5S ribosomal RNA
RF00002	152.2	74.8	5.8S ribosomal RNA
RF00005	73.0	63.7	tRNA
RF00008	55.2	75.8	Hammerhead ribozyme (type III)

The column of Len. means the average length of sequences, and the column of Homo. means the average homology of each pair of sequences calculated by the Smith–Waterman algorithm.

CRF and the EM algorithm are similar to each other in the point that both methods compute the expectation of the parameters with dynamic programming such as the inside–outside algorithm and optimize them by maximizing the likelihood of training data. However, the EM algorithm is for generative models, i.e. optimizes their parameters such that the model can most probably generate the training data, but has no guarantee to find the optimal alignments. However, CRF is for discriminative models, i.e. optimizes their parameters such that the model can most probably discriminate between correct alignments given by the training data and incorrect alignments. Therefore, for the task of finding the optimal alignment of given sequences, CRF is more suitable than the EM algorithm.

RIBOSUM is the matrix of log-odds scores of the probability distribution on RNA mutations obtained by the ML estimation against the background distribution on the RNA sequences, and has two meta-parameters: the first is the same as the BLOSUM matrices; the percentage of identical for grouping sequences, and the second is the lower bound of the homology between sequences which are used for calculating the matrix. Although these preprocessing of the dataset could help for avoiding overfitting, RIBOSUM still requires a large number of high quality structure-annotated alignments. In addition, there is no efficient method to determine the optimal meta-parameters because of a great variety of candidates for the two RIBOSUM meta-parameters along with some alignment parameters, such as gap penalties. However, CRF provides the efficient training algorithm that can determine the optimal parameter set and has the generalization ability so that a satisfiable score matrix can be obtained even with a small number of sample data without overfitting. In our experiments, CRF only uses ~100 structural alignments of RNA sequences for each RNA family and succeeds to obtain better score matrix than the others.

To investigate the effect of training data on the performance of structural alignment, we have compared three types of training data for each family: all the families, only the same family as the test data, i.e. the test data is a ‘seen’ family for the training data, and the other families excluding the test family, i.e. the test data is an ‘unseen’ family for the training data. Every training data contains 400 pairs of sequences. The result in Table 4 indicates that if the test data is a seen family for the training data, there are no significant differences in the accuracy of predicting base pairs in comparison with using all the families, and even if the test data is an unseen family for the training data, our method yet retains high accuracy. This result suggests that our method could align RNA sequences of unknown families with high accuracy.

We have done more practical experiments for predicting non-coding RNA regions by local alignment searches with a folded

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AA +0.20
AC -0.94 +2.10
AG -0.37 -0.46 -0.29
AU -1.10 +1.75 -1.02 +3.35
CA -0.42 -0.32 -0.24 -0.42 -0.36
CC +0.12 -0.39 -0.13 -0.71 -0.63 +0.18
CG +0.17 +0.23 +0.04 +2.27 +0.89 -0.47 +3.18
CU -0.32 +0.29 -0.35 +0.27 -0.39 -0.75 -0.93 +0.35
GA +0.08 -0.70 -0.25 -1.30 +0.23 -0.27 -0.17 -0.18 +0.44
GC -0.83 +1.18 -0.48 +2.72 +0.35 -0.44 +2.29 -1.08 -0.44 +3.15
GG -0.39 -0.45 -0.49 -0.60 -0.59 -0.09 +0.68 -0.21 -0.29 -0.86 -0.11
GU -0.31 -0.62 -0.88 +1.61 -0.61 -1.03 +0.82 -0.26 +0.95 +1.61 -1.44 +2.36
UA +0.14 -0.13 -0.26 +2.26 -0.49 +0.44 +2.26 -1.08 +0.18 +1.88 -0.74 -0.07 +2.37
UC -0.23 +1.21 -0.24 -1.23 -0.75 -0.33 -0.70 -0.28 -0.55 -0.77 -0.54 -1.01 +0.58 +0.55
UG -0.01 +0.47 -0.30 +0.96 -0.14 -1.14 +1.35 -0.60 -0.23 +1.35 -1.22 -1.05 +1.39 -0.45 -0.00
UU -0.64 -0.84 -0.29 +0.00 -0.24 -0.67 +0.58 -0.15 -0.78 -0.21 -0.58 -0.80 -0.68 -0.55 +0.84 +0.78
-- -1.40 -1.46 -1.28 +1.03 -1.23 -0.50 +0.93 -0.66 +0.27 +1.38 -1.57 +0.53 +0.67 -1.52 -0.51 -1.13
     AA    AC    AG    AU    CA    CC    CG    CU    GA    GC    GG    GU    UA    UC    UG    UU

 A +0.67
 C -1.11 -0.44
 G -1.02 -1.96 -0.52            M +2.25 -0.01 -0.55      M M M +0.002     A +0.45
 U -1.17 -1.30 -2.03 -0.68      I -1.83 -0.68 -1.38      M D M -0.002     C +0.63
 - +0.51 +0.19 +0.22 +0.25      D -1.53 -1.97 -0.72      M M D -0.002     G +0.49
    A     C     G     U            M     I     D         D D D -0.000     U -0.64
```

Fig. 2. The parameters of the weight vector $\boldsymbol{\lambda}$ corresponding to the emission features σ and the state transition features τ for structural alignment. The upper matrix is the parameters of the emissions of base pairs. The most left of the lower matrix is for the emissions of single bases. The second of the left is for the state transition $X \rightarrow Y$, where each column indicates X and each line Y. The third is for the state transitions with bifurcation. The most right of the lower matrix is for block insertions corresponding to the state transitions $I \rightarrow XI$ and $I \rightarrow IX$.

Table 3. The results of predicting base pairs by the proposed method (the columns of CRF) and the others

	CRF		EM		RIBOSUM		FOLDALIGN	
	SP (%)	SN (%)	SP (%)	SN (%)	SP (%)	SN (%)	SP (%)	SN (%)
RF00001	89.7	89.5	87.1	84.6	86.6	86.4	74.5	74.3
RF00002	84.7	82.5	85.6	76.6	79.5	76.3	55.0	51.7
RF00005	97.3	97.0	94.3	89.3	89.3	88.7	77.2	76.0
RF00008	91.5	92.2	89.5	80.4	91.0	91.2	87.2	87.0
Total	90.4	89.7	88.7	82.8	86.0	84.9	72.1	70.6

The columns of SP and SN indicate specificity and sensitivity of predicting base pairs the rate of correctly predicted base pairs to all of predicted base pairs, and the rate of correctly predicted base pairs to all of the annotated base pairs in the dataset, respectively.

Table 4. The difference of whether the test data is a seen family or an unseen family

	All families		Only the same family		The other families	
	SP (%)	SN (%)	SP (%)	SN (%)	SP (%)	SN (%)
RF00001	89.7	89.5	90.9 (+1.2)	91.4 (+1.9)	88.2 (−1.5)	87.9 (−1.6)
RF00002	84.7	82.5	87.3 (+2.6)	86.4 (+3.9)	78.7 (−6.0)	76.3 (−6.2)
RF00005	97.3	97.0	96.8 (−0.5)	96.8 (−0.2)	95.6 (−1.7)	95.7 (−1.3)
RF00008	91.5	92.2	90.1 (−1.4)	91.1 (−1.1)	86.6 (−4.9)	90.2 (−2.0)
Total	90.4	89.7	91.3 (+0.9)	91.4 (+1.7)	87.3 (−3.1)	87.5 (−2.2)

The columns of 'all families', 'only the same family' and 'the other families' indicate the family of the training data used for building the model. Each value in parentheses is the difference from the column of 'all families' on the same line.

sequence of the target RNA on genome sequences. The regions of DNA subsequences significantly aligned into the folded RNA structure are predicted as possible candidates of RNA coding regions of the same RNA family. For this experiment, we first prepare some longer DNA sequences including the true RNA coding sequences by appending dozens of nucleotides randomly with the same composition to the upstream and downstream of the RNA sequences. Second, we predict RNA regions on the prepared DNA sequences by aligning into the same family of a folded RNA structure. Then, we evaluate the correctness of the predicted RNA regions by counting errors which are the number of nucleotides not included in the region. Table 5 shows the average of prediction errors for each family. The column of PHMM indicates the prediction errors by the usual (unstructured) pair hidden Markov models. The result indicates

Table 5. The average of prediction errors of non-coding RNA regions

	CRF	EM	RIBOSUM	FOLDALIGN	PHMM
RF00001	5.06	6.24	7.67	35.95	45.83
RF00002	8.28	9.79	10.88	37.99	42.98
RF00005	7.32	8.72	10.20	36.15	45.31
RF00008	6.19	9.85	11.06	30.94	41.47
Total	6.71	8.65	9.95	35.26	43.90

that PHMM completely fails to predict RNA regions because of lack of ability to deal with any secondary structures, and structural alignment based on CRFs is more accurate for predicting non-coding RNA regions than the other scoring methods, such as EM, RIBOSUM and FOLDALIGN. These experimental results strongly support our discriminative method employing CRFs to estimate the score matrix parameters.

4 CONCLUSION

We have proposed a novel approach for RNA structural alignment based on conditional random fields (CRFs). Our approach has some specific features compared with previous methods in the sense that the parameters for structural alignment are estimated such that the model can most probably discriminate between correct alignments and incorrect alignments, and has the generalization ability so that a satisfiable score matrix can be obtained even with a small number of sample data without overfitting. Experimental results clearly show that the parameter estimation with CRFs can outperform all the other existing methods for structural alignment of RNA sequences. Furthermore, structural alignment search based on CRFs is more accurate for predicting non-coding RNA regions than the other scoring methods. These experimental results strongly support our discriminative method employing CRFs to estimate the score matrix parameters. This method will provide biologists explicit guidance for exploring non-coding RNA regions. In future, we would further improve the accuracy of predicting RNA secondary structures and non-coding RNA regions by employing more complex features for CRFs such as high-order dependencies of states, although only the first-order state transitions are used in our experiments.

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REFERENCES

Chen,S.F. and Rosenfeld,R. (1999) A Gaussian prior for smoothing maximum entropy models. *Technical Report CMU-CS-99-108*, Carnegie Mellon University,

Durbin,R. *et al*. (1998) *Biological Sequence Analysis*. Cambridge, UK, Cambridge University Press.

Eddy,S.R. (2001) Non-coding RNA genes and the modern RNA world. *Nat. Rev. Genet.*, **2**, 919–929.

Eddy,S.R. and Durbin,R. (1994) RNA sequence analysis using covariance models. *Nucleic Acids Res.*, **22**, 2079–2088.

Gorodkin,J. *et al*. (1997) Finding the most significant common sequence and structure motifs in a set of RNA sequences. *Nucleic Acids Res.*, **25**, 3724–3732.

Griffiths-Jones,S. *et al*. (2003) Rfam: an RNA family database. *Nucleic Acid Res.*, **31** 439–441.

Henikoff,S. and Henikoff,J.G. (1992) Amino acid substitution matrices from protein blocks. *Proc. Natl Acad. Sci. USA*, **89**, 10915–10919.

Klein,R.J. and Eddy,S.R. (2003) RSEARCH: finding homologs of single structured RNA sequences. *BMC Bioinformatics*, **4**, 44.

Lafferty,J. *et al*. (2001) Conditional random fields: probabilistic models for segmenting and labeling sequence data. In *Proceedings of the 18th International Conference on Machine Learning*, William College, MA, pp. 282–289.

Liu,D.C. and Nocedal,J. (1989) On the limited memory BFGS method for large-scale optimization. *Mathematical Programming*, **45**, 503–528.

Liu,Y. *et al*. (2004) Comparison of probabilistic combination methods for protein secondary structure prediction. *Bioinformatics*, **20**, 3099–3107.

Matsui,H. *et al*. (2005) Pair stochastic tree adjoining grammars for aligning and predicting pseudoknot RNA structures. *Bioinformatics*, **21**, 2611–2617.

McDonald,R. and Pereira,F. (2004) Identifying gene and protein mentions in text using conditional random fields. In *A critical assessment of text mining methods in molecular biology (BioCreative 2004)*, March 28–31, Granada, Spain.

Sakakibara,Y. (2003) Pair hidden Markov models on tree structures. *Bioinformatics*, **19**(Suppl. 1), S232–S240.

Sakakibara,Y. *et al*. (1994) Stochastic context-free grammars for tRNA modeling. *Nucleic Acids Res.*, **22**, 5112–5120.

Schattner,P. (2002) Searching for RNA genes using base-composition statistics. *Nucleic Acids Res.*, **30**, 2076–2082.

Settles,B. (2004) Biomedical named entity recognition using conditional random fields and rich feature sets. In *Proceedings of the COLING 2004 International Joint Workshop on Natural Language Processing in Biomedicine and its Application (NLPBA)*, August 28–29, Geneva, Switzerland, pp. 101–107.

Sha,F. and Pereira,F. (2003) Shallow parsing with conditional random fields. In *Proceedings of Human Language Technology-NAACL*, May 27–June 1, Edmonton Canada, pp. 213–220.